GCLS^2: Towards Efficient Community Detection Using Graph Contrastive Learning with Structure Semantics (Supplementary Material)

Anonymous Author(s)
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A Model Training

The detailed GCLS^2 training process is shown in Algorithm 2. Specifically, we first obtain the high-level structure graph G^H and high-level adjacency matrix A^H from the original data graph G, and construct a dictionary $\mathcal{D}$ by counting the number of specified patterns for each edge of G, then extract the structure similarity matrix S from the $\mathcal{D}$ (lines 1-3). For each training iteration, we first use the SSS module to extract structure semantics representation S' and attribute semantics representation X' (lines 4-6). Next, we employ GCNs with weight shared to extract node features from both G and G^H, resulting in the node feature matrix Z and the high-level node feature matrix Z^H (lines 7-8). Then, we calculate the similarity between positive samples and the similarity between negative samples to obtain the structure contrastive loss $(\ell(z_i^1), \ell(z_i^2))$ for each vertex v_i (line 9). After that, we update the GCLS^2 network parameters by minimizing all structure contrastive loss $\mathcal{L}$ on Z and Z^H (lines 10-11). Finally, after ε training iterations, we return the node features Z obtained for subsequent community detection (line 12).

B Complexity Analysis

We analyze the time complexity and space complexity of the GCLS^2 model by the number of floating point operations (FLOPs) and space overhead required by the GCLS^2 model. Specifically, let the time required for each unit floating point operation be $O(1)$. For the *Structure Similarity Semantic Encoder* module, the time complexity of the SSS module is $O(Nd^2)$, and the time complexity of the node representation extractor is $O(|E|d)$. For the *Structure Contrastive Learning* module, the similarity of each pair of vertices can be realized by vector dot product, which would cost $O(d)$. Then, the time complexity of all structure contrastive loss is $O(Nd)$. Therefore, the time complexity of the training process of GCLS^2 is $O(Nd^2 + |E|d)$.

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For the *Structure Similarity Semantic Encoder* module, the space complexity is $O(N^2 + d^2)$, which contains the input graph size and the SSS parameter size. Similarly, the space complexity of the node representation extractor is $O(Nd+d^2)$. For the *Structure Contrastive Learning* module, the computation of the structure contrastive loss can be performed on the feature matrices Z and Z^H with a space complexity of $O(N^2)$. Since the individual training process is serial and d is significantly smaller than N, the space complexity of the training process of GCLS^2 is $O(N^2)$.

For the HGP algorithm, the time complexity of the partitioning is $O(pN + |E|)$. The space complexity of the partitioning is $O(N)$.

C Theoretical Analyses

First, we have:

Lemma 1. *Let* $\mathbf{X}$ *be a variable with random uniform distribution after GNN coding, where* $\mathbf{X}_i$ *is the output vector of the node within the domain indicated by the GNN architecture. Given two embedding representations* $\mathbf{Z}^1, \mathbf{Z}^2 \in \mathbb{R}^L$ *of view 1 and view 2 are two random variables, their joint distribution is denoted as* $p(\mathbf{Z}^1, \mathbf{Z}^2)$. *Our objective* $\mathcal{L}$ *is a lower bound on the mutual information of* $\mathbf{X}$ *and* $(\mathbf{Z}^1, \mathbf{Z}^2)$ *as follows,*

$$-\mathcal{L} \le I(\mathbf{X}; \mathbf{Z}^1, \mathbf{Z}^2). \tag{1}$$

Proof. For a rigorous comparison, we compare our objective with the GRACE objective [1–3], which is defined under existing conditions as follows,

$$\Gamma(z_i^1) = \sum_{j=1}^{N} e^{sim(z_i^1, z_j^2)/\tau}, \tag{2}$$

$$\gamma(z_i^1) = \sum_{j=1, j\neq i}^{N} e^{sim(z_i^1, z_j^1)/\tau}, \tag{3}$$

$$I_{ACE}(\mathbf{Z}^1; \mathbf{Z}^2) \triangleq \mathbb{E}_{\prod_i p(z_i^1, z_i^2)} [\frac{1}{N} \sum_{i=1}^{N} log \frac{e^{sim(z_i^1, z_i^2)/\tau}}{\Gamma(z_i^1) + \gamma(z_i^1)}], \tag{4}$$

where $\prod_i p(z_i^1, z_i^2)$ denotes the joint distribution. To facilitate with the notational representation, we define $\Gamma(z_i^1) = \sum_{j=1}^{N} e^{sim(z_i^1, z_j^2)/\tau}$, $\gamma(z_i^1) = \sum_{j=1, j\neq i}^{N} e^{sim(z_i^1, z_j^1)/\tau}$. Please note that $\Gamma(z_i^2)$ and $\gamma(z_i^2)$ are defined symmetrically with $\Gamma(z_i^1)$ and $\gamma(z_i^1)$ respectively. So that our objective $\mathcal{L}$ can be rewritten as

$$-\mathcal{L} = \mathbb{E}_{\prod_i p(z_i^1, z_i^2)} [\frac{1}{N} \sum_{i=1}^{N} log \frac{pos(z_i^1) + e^{sim(z_i^1, z_i^2)/\tau}}{\sqrt{(\Gamma(z_i^1) + \gamma(z_i^1))(\Gamma(z_i^2) + \gamma(z_i^2))}}], \tag{5}$$

where the $\sqrt{(\Gamma(z_i^1)+\gamma(z_i^1))(\Gamma(z_i^2)+\gamma(z_i^2))}$ is to illustrate the symmetry of $\Gamma(z_i^1)+\gamma(z_i^1)$ and $\Gamma(z_i^2)+\gamma(z_i^2)$, which are equal in value. Therefore, we can use the Eq. 4 Eq. 5 to deduce that

$$\begin{aligned}-2\mathcal{L} &= I_{ACE}(\mathbf{Z}^1;\mathbf{Z}^2) - \mathbb{E}_{\prod_i p(z_i^1,z_i^2)}\left[\frac{1}{N}\sum_{i=1}^{N} log(1+\frac{pos(z_i^1)}{e^{sim(z_i^1,z_i^2)/\tau}})\right] \\ &+ I_{ACE}(\mathbf{Z}^2;\mathbf{Z}^1) - \mathbb{E}_{\prod_i p(z_i^1,z_i^2)}\left[\frac{1}{N}\sum_{i=1}^{N} log(1+\frac{pos(z_i^2)}{e^{sim(z_i^2,z_i^1)/\tau}})\right] \\ &\le I_{ACE}(\mathbf{Z}^1;\mathbf{Z}^2) + I_{ACE}(\mathbf{Z}^2;\mathbf{Z}^1).\end{aligned} \tag{6}$$

And according to [2, 3], the GRACE objective is a lower bound of the true MI, i.e.,

$$I_{ACE}(\mathbf{Z}^1;\mathbf{Z}^2) \le I(\mathbf{Z}^1;\mathbf{Z}^2). \tag{7}$$

Thus, we have the following inference

$$-2\mathcal{L} \le I(\mathbf{Z}^1;\mathbf{Z}^2) + \le I(\mathbf{Z}^2;\mathbf{Z}^1) \le 2I(\mathbf{Z}^1;\mathbf{Z}^2). \tag{8}$$

This is equivalent to

$$-\mathcal{L} \le I(\mathbf{Z}^1;\mathbf{Z}^2). \tag{9}$$

For three random variables $\mathbf{X}$, $\mathbf{Y}$, $\mathbf{Z}$ satisfying one Markov chain $\mathbf{X} \to \mathbf{Y} \to \mathbf{Z}$, we have $I(\mathbf{X};\mathbf{Z}) \le I(\mathbf{X};\mathbf{Y})$ holds. We find that $\mathbf{X}$, $\mathbf{Z}^1$, $\mathbf{Z}^2$ satisfy the chain relation $\mathbf{Z}^1 \leftarrow \mathbf{X} \to \mathbf{Z}^2$. Actuality, $\mathbf{Z}^1$ and $\mathbf{Z}^2$ are conditionally independent after considering $\mathbf{X}$. This represents the existence of Markov chain $\mathbf{Z}^1 \to \mathbf{X} \to \mathbf{Z}^2$, and we have $I(\mathbf{Z}^1;\mathbf{Z}^2) \le I(\mathbf{Z}^1;\mathbf{X})$. Since the chain relation $\mathbf{X} \to (\mathbf{Z}^1,\mathbf{Z}^2) \to \mathbf{Z}^1$ holds, it follows that $I(\mathbf{X};\mathbf{Z}^1) \le I(\mathbf{X};\mathbf{Z}^1,\mathbf{Z}^2)$. Combining the two inequalities above, we have the inequality

$$I(\mathbf{Z}^1;\mathbf{Z}^2) \le I(\mathbf{X};\mathbf{Z}^1,\mathbf{Z}^2). \tag{10}$$

According to Eq. 9 and Eq. 10, we can derive the final inequality as follows and complete the proof,

$$-\mathcal{L} \le I(\mathbf{X};\mathbf{Z}^1,\mathbf{Z}^2). \tag{11}$$

□

Analysis: The mutual information $I(\mathbf{X};\mathbf{Z}^1,\mathbf{Z}^2)$ measures the information content of the original data $\mathbf{X}$ retained in the learned representations $\mathbf{Z}^1$ and $\mathbf{Z}^2$. From Lemma 1, the optimization of the contrastive loss $\mathcal{L}$ directly relates to the lower bound of this mutual information. Specifically, as $\mathcal{L}$ decreases (i.e., the model becomes better optimized), the lower bound of the mutual information $I(\mathbf{X};\mathbf{Z}^1,\mathbf{Z}^2)$ increases. This mechanism prevents excessive loss of original data information in representation $\mathbf{Z}$, thereby enhancing the performance of downstream community detection tasks.

D Experimental Evaluation Supplement

HGP Performance Evaluation Figure 1 presents the performance evaluation of the HGP algorithm with different partition size p. From Figure 1, when the partition size $p = 32$, the HGP algorithm completes the partitioning of graphs with tens of millions of edges in just 155.91 seconds. Combining these findings with Table 4, it is clear that the GCLS^2+HGP method performs well in terms of efficiency and accuracy of community detection in large-scale datasets.

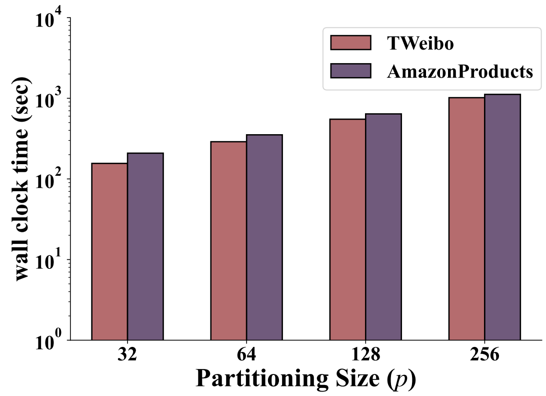

Figure 1: The partitioning cost time comparison of different partition size on large-scale graph datasets.

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